

Vehicle Rental System

A Java and SQL Based Object-
Oriented Project

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Certificate

- Certified that the project entitled 'Vehicle Rental System' is a bonafide record of work done by:
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- in partial fulfillment of the requirements for the Object-Oriented Programming course.

Acknowledgment

- We would like to express our gratitude to our faculty and peers for their guidance and support throughout the completion of this project. Their insights helped us in understanding Java, SQL, and software development principles effectively.

Abstract

- The Vehicle Rental System is designed to automate the process of renting and managing vehicles. It allows customers to view available vehicles, book rentals, and maintain records using a database. The system ensures efficiency, accuracy, and persistent storage through Java and SQL integration.

Introduction

- The project demonstrates Object-Oriented Programming principles using Java and SQL. It enables users to rent, return, and manage vehicles efficiently. The system replaces manual processes with an automated digital solution.

Problem Statement

- Manual vehicle rental management causes errors, data loss, and difficulty in tracking rentals. There is a need for a digital solution that offers efficient data management, faster operations, and accurate records.

Objectives

- • Automate the process of renting and managing vehicles
- • Store and retrieve data using SQL
- • Reduce manual errors and enhance user experience
- • Implement OOP concepts for modular and reusable code

Scope of the Project

- The system can be extended for use in car, bike, or equipment rentals. It can include multiple user roles (admin, customer) and be integrated with online payment gateways and tracking systems.

System Requirements

- Software Requirements:
 - • Java JDK 11 or above
 - • MySQL / SQLite
 - • JDBC Connector
 - • Eclipse or IntelliJ
- Hardware Requirements:
 - • Minimum 4GB RAM
 - • 500MB storage
 - • Windows / Linux OS

System Architecture

- Frontend: Java Console Application
- Middleware: JDBC (handles database connection)
- Backend: SQL Database (stores vehicle, rental, and user data)
- Data Flow: User → Java Program → JDBC → SQL Database → Program → User

Module Description

- Modules implemented:
- • Vehicle Management – Add, view, delete vehicles
- • Rental Management – Rent and return vehicles
- • Customer Management – Handle customer records
- • Transaction Module – Maintain rental history

Class Design (OOP Concepts)

- Classes:
 - Vehicle – Stores vehicle ID, model, type, availability
 - Customer – Holds customer details
 - RentalTransaction – Tracks each rental record
 - VehicleRentalSystem – Main control class

Database Design

- Tables:
 - Vehicles (vehicle_id, model, type, status)
 - Customers (cust_id, name, contact)
 - Rentals (rental_id, vehicle_id, cust_id, rent_date, return_date)
- Relationships:
 - One customer can have multiple rentals
 - One vehicle can be rented multiple times

Implementation

- Language: Java
 - Database: MySQL/SQLite
 - Connectivity: JDBC
 - IDE: Eclipse or IntelliJ
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- Functions:
 - • Add Vehicle
 - • View Vehicle
 - • Rent Vehicle
 - • Return Vehicle
 - • Display Records

Input and Output

- Input: Vehicle details, customer info, rental dates
- Output: Rental status, available vehicles, rental history
- The system displays all interactions in a user-friendly menu-based console interface.

Testing and Validation

- Testing ensured all modules worked correctly. Functional and unit testing confirmed:
 - • Accurate data storage and retrieval
 - • Successful connection between Java and SQL
 - • Correct handling of user input and errors

Results and Discussion

- The project successfully automated vehicle rental operations. All data was stored persistently, and transactions could be viewed or updated efficiently. The system improved accuracy and speed.

Outcomes

- • Efficient rental process
- • Organized data storage
- • Reduced human errors
- • Easy search and tracking
- • Scalable for future enhancements

Future Enhancements

- • Add Graphical User Interface (GUI)
- • Integrate payment and GPS tracking
- • Add authentication for admin and users
- • Generate monthly reports and analytics

Conclusion & References

- The Vehicle Rental System efficiently automates rental management using Java and SQL.
- It demonstrates OOP, JDBC, and data handling skills.
- References:
 - • Java Documentation
 - • JDBC API Docs
 - • SQL Reference Manual
 - • Online Java Tutorials